Foundation Models for Automated Design of Energy Flow Systems

Budget Justification - University of Michigan

Salaries and Wages:

Senior/Key Personnel

Support in the amount of $16,667 is requested for the Principal Investigator (PI) Ricardo Vinuesa for 100% of 1 month summer effort in Year 1 of the nine-month project. The current base rate for Dr. Vinuesa is $150,000 for the 2025-2026 academic year (AY) and anticipates a 3% increase per academic year, effective July 1st of each year.

The University of Michigan uses the following definition for the purpose of calculating the two-month limit on salary requested in the proposal budget for senior personnel:

- A fiscal year is the 12-month period starting July 1 and ending June 30.

Other Personnel:

Graduate Student Research Assistant

Support in the amount of $26,309 is requested for a Graduate Student Research Assistant (GSRA). The GSRA will have an appointment of 50% from 9/1/2026 – 3/31/2027. A 50% appointment is considered full-time as the other 50% of their time is dedicated to studies and other academic pursuits. The current base monthly rate for a GSRA is $7,298. Please refer to https://hr.umich.edu/sites/default/files/2025-2026_gsa_salary_memo_ann_arbor_7.7.25.pdf for The University of Michigan's GSRA monthly rate.

Fringe Benefits:

Fringe benefits are estimated at 30% of salaries and wages for the PI. This is an average rate. The fringe benefits for the GSRA are estimated at 20% of wages. The project will be charged only those actual costs for fringe benefits of personnel assigned to the project. Estimated fringe benefits total $10,268. Please refer to http://www.orsp.umich.edu/proposals/budgets/staff_benefits.html for additional information on how fringe benefits are estimated.

Tuition:

The GSRA compensation package includes tuition in the amount of $17,736 for 1.75 semesters of tuiotion. The GSRA is expected to be candidate status during the project. Tuition is prorated for the last term as the end date falls in the middle of the semester. The current base rate for a candidate status Ph.D. student is $9,652 per term based on an in-state, average cost for a full-time Ph.D. student. An increase of 5%, effective September 1st of each year, has been carried through the calculations for each year of the proposal. For additional details concerning The University of Michigan's Graduate Student Research Assistants please see:

Pre-candidate tuition: http://www.orsp.umich.edu/proposals/students/gsra.html

Candidate Tuition: https://ro.umich.edu/tuition-residency/tuition-fees?year=140&school=93&term_type=75&level=84

Domestic Travel:

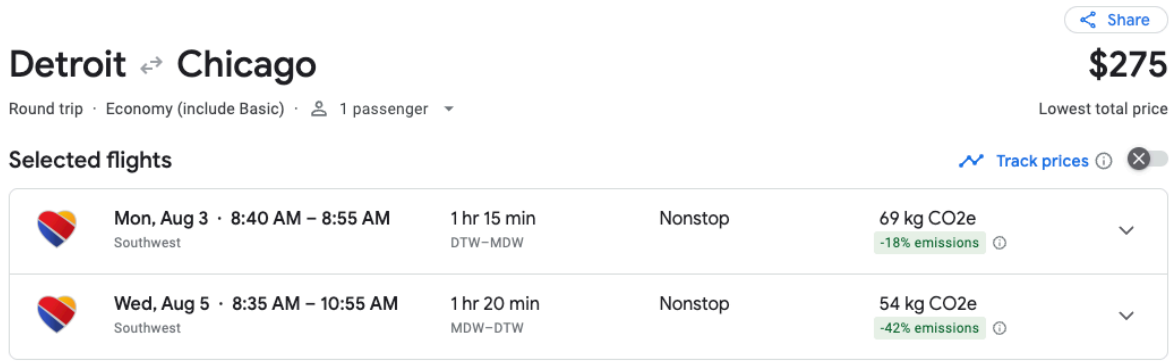
Travel support of $2,400 is requested for the PI to travel to Argonne National Laboratory participate in coordination activities with other projects two times. Costs may include airfare, lodging, basic meals, and ground transportation related costs.

All lodging and M&IE (meals and incidental expense) amounts are estimated by using current General Services Administration (GSA) rates and can be found at: https://www.gsa.gov/travel/plan-book/per-diem-rates. All other estimated travel expenses are based on historic data from similar travel, a sample

airfare quote, and 2026 SciTech registration costs. Only actual travel charges will be charged to the project. The following is a breakdown of estimated travel expenses.

Biannual Project review meetings, 3-night trip for PI to Argonne National Laboratory in Lemont, IL			
	Subtotal	Multiplier	Line Total
Airfare (origin: DTW)	275	1	275
Per Diem (75% on first and last days)	80	3.5	280
Ground Transportation	75	4	300
Lodging	115	3	345
3-night trip for PI		**Total per trip:**	**$ 1,200**

Sample Flight Itinerary and Price:



Computing Supplies
Funds are budgeted in the amount of $3,500 for the purchase of a laptop, workstation, and hardware components such as hard drives and cables. The computers and associated supplies will be used by the PI and GSRA for project related tasks such as, conducting research, documenting data and project findings, and communicating with various project stakeholders. The computers and associated supplies will be used exclusively for this project. Estimated computer costs are based on historic data from similar type projects.

Facilities and Administrative Costs:
Per the negotiated agreement with the Federal Government, beginning May 22nd, 2025, the indirect cost rate is 56% on Modified Total Direct Costs (MTDC), that is, Total Direct Costs less the cost of student tuition, equipment, and individual subcontractor costs greater than $25,000. Estimated facilities and administrative costs total $33,121. For additional information on the University of Michigan's Indirect Cost rate please refer to: http://orsp.umich.edu/proposals/budgets/indirect_cost_rates.html.
Federal Agency is the Department of Health and Human Services, Olulola Oluborode (301) 492-4065, olulola.oluborode@psc.hhs.gov

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operate federally supported research and programs in compliance with all applicable grant terms and conditions, consistent with current laws, regulations, and related court decisions. The University regularly examines programs, policies, and activities for compliance with university policy and federal and state law. The University's ongoing compliance processes include monitoring, audits, a compliance hotline, and training. The University takes prompt corrective action when it has direct knowledge of non-compliance with policy or applicable law.